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100+

AI Engineering Concepts

I Want to Explore

A roadmap, not a checklist.

LLM Architecture

- Transformer Architecture
- RoPE & ALiBi
- GQA & MQA
- FlashAttention
- PagedAttention
- Sparse & Linear Attention
- Mamba
- Mixture of Experts (MoE)
- Speculative Decoding
- Test-Time Scaling

LLM Inference

- Continuous Batching
- Dynamic Batching
- Chunked Prefill
- Prefix Caching
- Semantic Caching
- KV Cache
- Token Throughput
- TTFT
- CUDA Graphs
- Kernel Fusion

GPU & Systems

- GPU Architecture
- CUDA Memory Hierarchy
- Tensor Cores
- VRAM Optimization
- PCIe vs NVLink
- GPUDirect
- NCCL
- RDMA
- CUDA Streams
- Warp Scheduling

Quantization & Fine-Tuning

- INT8
- FP8
- GPTQ
- AWQ
- GGUF
- EXL2

- LoRA
- QLoRA
- DoRA
- DPO
- RLHF
- Knowledge Distillation

Retrieval Systems

- Agentic RAG
- Self-RAG
- CRAG
- GraphRAG
- LightRAG
- HippoRAG
- RAPTOR
- ColBERT
- HyDE
- DSPy
- Context Compression
- Semantic Routing

AI Agents

- ReAct
- Reflexion
- Tree of Thoughts
- Graph of Thoughts
- Tool Calling
- Agent Memory
- Multi-Agent Systems
- Agent Orchestration
- Planning Algorithms
- Computer Use Agents

Serving & Infrastructure

- vLLM
- SGLang
- TensorRT-LLM
- llama.cpp
- Triton Inference Server
- Ray Serve
- BentoML
- Ollama
- KServe
- Distributed Inference

Distributed Training

- DDP
- FSDP
- ZeRO
- Tensor Parallelism
- Pipeline Parallelism
- Sequence Parallelism
- Context Parallelism
- Expert Parallelism
- DeepSpeed
- Megatron-LM

Evaluation & Context Engineering

- Langfuse
- Ragas
- TruLens
- Promptfoo
- LLM-as-a-Judge
- Hallucination Detection

- MCP
- Prompt Compression
- Prompt Routing
- Context Windows
- Structured Outputs

Research Frontiers

- Mechanistic Interpretability
- Sparse Autoencoders
- Circuit Analysis
- MEMIT
- ROME
- Scaling Laws
- Chinchilla Scaling
- Small Language Models
- Diffusion Language Models
- World Models
- AI Compiler Stack
- Neural Memory

AI System Design

- AI System Design
- Model Routing
- Multi-Model Orchestration
- LLM Gateway Architecture
- Cost-Aware Inference
- Latency-Aware Routing
- Guardrails
- Prompt Injection
- AI Security
- Production AI Architecture

There has never been a better time to be curious.

The AI stack is evolving incredibly fast, and there's always something new to learn.

What topic would you dive into first?

Hishita Gupta